Diego Gerardo Sánchez Moreno

Software/Product Engineer — Co-founder & Project Director, Autónoma (AutoNomaMX)
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Summary

Software and product engineer who ships end to end: problem, design, database, security, deployment and daily operation. At a private school, sole author of two web systems in production (SATAG and SEVAD) and a cryptographic prototype (SIVEA); co-founder & Project Director at Autónoma (AutoNomaMX), with three point-of-sale systems in production on a multi-tenant backend, delivered with formal project management (PMBOK). Strongest in PostgreSQL, security and privacy. Full portfolio with projects and demos: diegogerardo.com.

Education

B.S. in Robotics and Digital Systems Engineering
Tecnológico de Monterrey, Campus Querétaro — 2022 – 2026
Professional license (Cédula Profesional) 15796458 — SEP, Mexico, 2026

Experience

Autónoma (AutoNomaMX) — Co-founder & Project Director
Querétaro — Jan 2026 – Present

- Shipped three offline-first POS systems now in production with real clients (a restaurant, a coffee shop, and an optical shop) on a multi-tenant PostgreSQL/Supabase backend with Row-Level Security.
- Built a SaaS analytics platform with an AI assistant mediated by a semantic layer so reported figures stay accurate (Next.js, Stripe).
- Implemented CI/CD with self-hosted GitHub Actions runners on the production hardware (Raspberry Pi, Ubuntu): automated deploys with smoke tests on every push.
- Directed Phase 1 of the optical-shop project under PMBOK: charter, WBS, CPM, delivery with signed acceptance, client training, and closure with $0 cost deviation.
- Integrated POS hardware: ESC/POS thermal printers (USB and network) and Code128 barcode labels on Brother DK-1204.

Instituto Asunción de Querétaro — Software Developer / IT
Jun 2026 – Present

- Sole author of **SATAG**, a vehicle-access platform in production (public request form with on-screen electronic signature and SHA-256 evidence, cashier, inventory and parking analytics): 406 commits and 105 numbered, reversible SQL migrations; 40 tables all under Row-Level Security, 46 RLS policies, 78 functions and 104 indexes.
- Integrated a third-party access-control system with no API: an idempotent importer that read 1.5M log rows across 39 exports and de-duplicated 266k overlapping rows, plus a batch change engine (102 changes requested, 73 verified against the roster, 0 mismatches).
- Cut the parking dashboard from 41s to ˜6s and a name cross-match from 20.4s to 1.9s by profiling in production at each step.
- Shipped **SEVAD**, an anonymous QR service-evaluation system, from zero to production in 6 days; later optimized the dashboard from 28.9s to 23ms (diagnosed an N+1 and missing indexes with EXPLAIN) and fixed an admin-function exposure caused by PostgreSQL default privileges.
- Prototyped **SIVEA**, secure student pickup using rotating Ed25519 credentials whose private key never leaves the phone, with a native mobile app (Expo), property-based tests and a threat model.
- Work with AI-assisted development (Claude Code) under automatic controls: mandatory CI, an anti-pattern guard, post-deploy verification and an MCP integration to project management. Architecture, security, legal and operational decisions are mine.

Patio Cinco (Querétaro) — Production Platform
Product/Systems Engineer — 2025 – 2026

- Delivered a production system for memberships, billing, and customer operations with persistent fiscal data (Wix + Velo).

Hybrid Soft Robotics Lab (Tec de Monterrey, Qro.)
Engineering Research — 2024 – Present

- Built a telemetry and control platform with a Python SDK + GUI and MATLAB-based analysis.
- Implemented embedded communication (ESP32 + micro-ROS) and data logging pipelines.

Awards & Certifications

- Project Management program — Tecnológico de Monterrey (2026): Fundamentals (integrative exam), Scope, Cost, Time & Resources, Execution/Monitoring & Control
- Best Poster Award — Exploring Soft Robotics (Dec 2025)
- NVIDIA DLI: Fundamentals of Deep Learning (Jun 2025)
- OpenCV Bootcamp (Apr 2025)

Technical Skills

- **Languages:** TypeScript, Python, C++, SQL
- **Frontend:** Next.js, React, Astro, Tailwind CSS
- **Backend/Data:** FastAPI, Supabase/PostgreSQL (Row-Level Security), SQLite, Stripe, REST
- **Infra/Reliability:** Docker, CI/CD (GitHub Actions self-hosted runners), offline-first (Dexie/IndexedDB, SQLite queues), Raspberry Pi

Languages

Spanish (Native), English (B2; technical documentation and technical conversations)